Real time Stuttering Detection System

(AN AI-POWERED PLATFORM FOR ENHANCED CHILDERN'S LEARNING)

Project Final Report

I.Ilangeeran-IT22153104

B.Sc. (Hons) in Information Technology Specializing in Information Technology

Department of Information Technology

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DECLARATION

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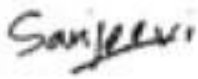
Name	Student ID	Signature
I.Ilangeeran	IT22153104	

The above candidate is carrying out research for the undergraduate dissertation under my supervision.



…… 24/4/2026

Signature of the supervisor.....Date

(Ms.Sanjeevi Chandrasiri)

…… 24/4/2026

Signature of co-supervisor.....Date

(Ms.Karthiga Rajendran)

ABSTRACT

Stuttering is a speech disorder that disrupts the normal flow of speech through repetitions, prolongations, or involuntary pauses. Early identification of stuttering is important for effective speech therapy and communication development. However, traditional detection methods mainly rely on manual evaluation by speech therapists, which can be time-consuming, subjective, and not always accessible to individuals who require early screening. This project proposes a real-time stuttering disorder detection system using machine learning techniques to automatically analyze speech patterns and classify speech fluency.

The system accepts speech input either from recorded audio files or through a live microphone. The audio signal first undergoes preprocessing to remove noise and normalize the signal for consistent analysis. Key acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and zero-crossing rate are extracted using audio signal processing techniques. These features are then standardized and used as input for a Random Forest classification model trained to distinguish between normal speech and stuttering speech patterns.

To evaluate the reliability of the model, K-Fold cross validation was applied during training, achieving an overall classification accuracy of approximately 93 percent. The system provides prediction results along with confidence scores and categories detected stuttering into different severity levels to assist in further analysis.

The proposed system demonstrates how machine learning and speech signal processing can be combined to support automated stuttering detection. This approach has the potential to assist speech therapists, enable early screening, and provide a foundation for future intelligent speech assessment tools.

ACKNOWLEDGEMENT

The success of this research would not have been possible without the support and guidance of many individuals.

First, I would like to express my sincere gratitude to my supervisor, Dr. Sanjeevi Chandrasiri, and co-supervisor, Ms. Karthiga Rajendran, for their continuous guidance, encouragement, and valuable advice throughout the research project. Their expertise and constructive feedback were essential in completing this work successfully.

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My heartfelt appreciation goes to my group members for their cooperation, teamwork, and assistance during the research process.

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LIST OF ABBREVIATIONS

Abbreviations	Description
AI	Artificial Intelligence
ML	Machine Learning
MFCC	Mel-Frequency Cepstral Coefficients
ZCR	Zero Crossing Rate
RF	Random Forest
DT	Decision Tree
K-Fold CV	K-Fold Cross Validation
DSP	Digital Signal Processing
API	Application Programming Interface

I. INTRODUCTION

1.1 Background Study and Literature Review

1.1.1 Background Study

Stuttering is a speech disorder that affects the natural flow of speech through repetitions, prolongations, and pauses. It can impact communication, confidence, and social interaction, especially if not identified early. Traditionally, stuttering detection relies on speech therapists who manually analyze speech patterns. However, this process can be time-consuming and may not always be accessible to everyone. With advancements in machine learning and speech processing, automated systems can now analyze speech signals and identify stuttering patterns more efficiently, supporting early detection and assisting professionals in speech therapy.

Figure 1: Child with stuttering Disorder

Speech disorders such as stuttering can vary in severity from mild interruptions to frequent disruptions in speech fluency. Early identification plays a crucial role in helping individuals receive appropriate therapy and support. In many cases, individuals may not have easy access to professional speech evaluation due to geographical limitations, lack of specialists, or high consultation costs. As a result, there is a growing interest in developing technological solutions that can assist in the early screening of speech disorders.

Recent developments in digital signal processing and artificial intelligence have made it possible to analyze complex speech patterns using computational techniques. Speech signals contain various acoustic characteristics such as pitch, frequency, energy, and temporal variations that can reveal irregularities in speech production. By extracting these features and analyzing them with machine learning algorithms, systems can learn to distinguish between normal speech and speech that contains stuttering patterns.

Machine learning models are particularly useful because they can be trained using labeled speech datasets and can identify hidden patterns that may not be easily detected through manual observation. Once trained, these models can quickly process new speech samples and provide classification results. Integrating these models with real-time audio processing allows the development of systems capable of providing immediate feedback.

Therefore, developing an automated stuttering detection system can contribute to improving early screening methods, supporting speech therapists, and making speech analysis tools more accessible for educational and healthcare environments.

Figure 4:System Overview

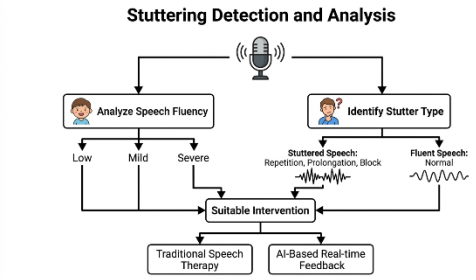
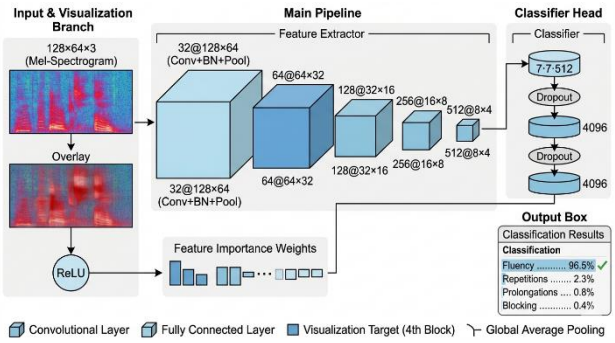


Figure 2:MFCC Audio prediction



1.2 Literature Review

Speech disorders, particularly stuttering, have gained increasing attention in the field of Information Technology, especially with the advancement of Artificial Intelligence (AI) and speech signal processing techniques. Researchers have explored various approaches to automatically detect and analyze stuttering patterns, aiming to reduce dependency on manual assessment by speech therapists. This section reviews related works that guided the development of the proposed system.

In recent years, machine learning and deep learning techniques have been widely applied in speech disorder detection. Howell et al. [1] demonstrated the use of acoustic feature analysis to identify stuttering events, showing that automated systems can effectively detect disfluencies such as repetitions and prolongations. Similarly, Alharbi et al. [2] applied machine learning classifiers, including Support Vector Machines (SVM) and Random Forest, to classify speech samples as fluent or disfluent, achieving promising accuracy. These studies highlight that feature-based machine learning models are effective for stuttering detection tasks.

Apart from classification, researchers have also focused on extracting meaningful speech features. According to Othman et al. [3], features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and zero-crossing rate play a crucial role in identifying speech abnormalities. These features help capture both temporal and spectral characteristics of speech signals, improving the performance of classification models. In another study, Wiseman and Rudzicz [4] explored the use of deep learning models such as Recurrent Neural Networks (RNNs) to analyze temporal dependencies in speech, further enhancing detection accuracy.

Another important area of research is severity assessment of stuttering. Traditional approaches rely on manual evaluation by speech therapists, which can be subjective and time-consuming. A study by Hariharan et al. [5] proposed an automated system that measures the frequency and duration of disfluencies to classify severity levels. This approach supports objective and consistent evaluation, which is essential for effective therapy planning.

In addition, several studies have explored real-time and user-friendly applications. Alshammari et al. [6] developed a web-based speech analysis tool that allows users to upload audio and receive instant feedback. However, many existing systems focus only on detection and lack integrated features such as severity analysis and therapy recommendations.

From the reviewed literature, it is evident that machine learning and speech processing techniques play a significant role in stuttering detection. While many studies have demonstrated accurate classification of speech disfluencies, only limited systems provide a complete pipeline that includes detection, severity analysis, and therapy guidance in a single platform. The proposed Real-Time Stuttering Disorder Detection System builds upon these works by integrating audio preprocessing, machine learning-based classification, severity assessment, and personalized therapy recommendations within a web-based application, making it more practical and accessible for real-world use.

1.3 Research Gap

Speech disorders such as stuttering continue to affect many individuals worldwide, influencing their communication abilities, social confidence, and overall quality of life. Early identification of stuttering is essential for providing timely speech therapy and intervention. Although several international studies have explored the use of machine learning and speech signal processing techniques to detect stuttering patterns, many of these solutions are developed in controlled laboratory environments using large and highly curated datasets. As a result, they are often designed for research purposes rather than practical, real-time applications that can be easily used in everyday environments.

Existing speech analysis tools often focus on general speech recognition or transcription tasks rather than specifically identifying fluency disorders such as stuttering. Many available systems also lack the capability to perform real-time speech analysis or provide immediate feedback based on speech irregularities. Furthermore, some research studies concentrate mainly on feature extraction or offline model training without integrating the complete pipeline required for a practical detection system.

Therefore, there is a clear research gap in developing an integrated system that combines real-time speech input, acoustic feature extraction, and machine learning–based classification to automatically detect stuttering patterns and provide meaningful feedback. This study addresses this gap by developing a real-time stuttering disorder detection system that utilizes speech signal processing techniques and a machine learning model to classify speech fluency and support early screening of stuttering.

Feature / Capability	Research [1]	Research [2]	Research [3]	Research [4]	Proposed System
Stuttering detection using ML/DL				+	
Severity detection of stuttering	+		+	+	
Therapy recommendation based on severity	+	+		+	
Real-time speech analysis (live input)	+	+	+		
Web-based application for user access	+	+	+	+	

Table 1 Research Gap

1.4 Research Problems

Speech fluency disorders such as stuttering affect many individuals and can significantly impact communication, confidence, and social interaction. Early detection of stuttering is important for providing effective speech therapy and improving communication skills. However, the current process of identifying stuttering mainly depends on manual evaluation by speech-language therapists. This traditional method presents several challenges:

- Subjectivity**:** Manual assessment may vary depending on the therapist's experience and interpretation of speech patterns.
- Limited accessibility**:** Access to professional speech therapists may be restricted due to geographical limitations or high consultation costs.
- Time-consuming evaluation**:** Continuous observation and analysis of speech recordings require significant time and effort.
- Lack of automated tools**:** Most existing speech analysis applications focus on speech recognition or transcription rather than detecting fluency disorders such as stuttering in real time.

Thus, the research problem can be defined as:

"How can a machine learning–based stuttering detection system be designed to accurately

analyze speech patterns, identify stuttering characteristics, and provide real-time classification of speech as normal or stuttering?

This research problem highlights the need for a solution that is accurate, efficient, and accessible, enabling automated speech analysis that can assist speech therapists and support early identification of stuttering disorders.

1.5 Research Objectives

Main Objective

To develop a machine learning–based real-time stuttering disorder detection system that analyzes speech signals, identifies stuttering patterns, and classifies speech fluency in order to support early detection and assist speech therapy processes.

Specific Objectives

1. To design and implement a machine learning model for classifying speech as normal or stuttering based on extracted acoustic features.
2. To extract important speech features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and zero-crossing rate using speech signal processing techniques.
3. To develop a real-time speech analysis module that accepts microphone or recorded audio input and processes it for stuttering detection.
4. To implement a severity analysis mechanism that categorizes detected stuttering into levels such as mild, moderate, and severe.
5. To evaluate and validate the performance of the model using techniques such as cross-validation and accuracy measurement to ensure reliable detection results.

Business Objectives

1. To assist speech therapists by providing an automated tool that can support the preliminary screening of stuttering disorders.
2. To enable early detection of speech fluency issues, allowing individuals to seek timely therapy and intervention.
3. To provide an accessible and user-friendly speech analysis system that can be used for monitoring speech patterns.
4. To contribute to the development of intelligent healthcare technologies by integrating machine learning and speech signal processing for speech disorder analysis.

2. METHODOLOGY

2.1 Methodology

The development of the Real-Time Stuttering Disorder Detection System followed a structured methodology to ensure accurate speech analysis, reliable stuttering classification, and effective real-time performance. The project began with data collection, where speech recordings containing both normal speech and stuttering speech patterns were gathered from publicly available speech datasets and research repositories. Each audio sample in the dataset was labeled according to speech fluency, categorizing the recordings as either normal speech or stuttering speech. These labeled samples were used to train the machine learning model to recognize patterns associated with speech disfluencies.

After data collection, audio preprocessing was performed to prepare the recordings for feature extraction. Background noise and silent segments were reduced to improve signal quality. The audio signals were then normalized to maintain consistent amplitude levels across all samples. This preprocessing stage ensured that the speech data was clean and suitable for further analysis.

Next, feature extraction was carried out using speech signal processing techniques. Important acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and zero-crossing rate were extracted using the Librosa library. These features represent important characteristics of human speech and help capture variations in speech fluency. The extracted features were then standardized using StandardScaler to ensure that all feature values were within a consistent range for effective machine learning model training.

For model development, a Random Forest classification model was selected to classify speech as either normal or stuttering. The dataset was divided into training and testing sets to evaluate the performance of the model. During training, the model learned patterns and relationships between the extracted features and their corresponding speech labels. K-Fold cross validation was applied to ensure reliable model performance and prevent overfitting. Performance metrics such as accuracy, precision, recall, and F1-score were used to evaluate the model's classification capability.

Finally, the system underwent evaluation and testing to validate its performance. Model predictions were compared with labeled speech samples to verify classification accuracy. The system was also tested with different speech recordings to ensure stable performance under varying speech patterns and speaking speeds. Usability testing was conducted to confirm that the system could operate efficiently in real-time conditions. Overall, this methodology provides a complete workflow that integrates data collection, preprocessing, feature extraction, model training, system integration, and performance evaluation to develop an effective real-time stuttering disorder detection system.

2.1.1 Agile Principles Applied in the Project

The development of the Real-Time Stuttering Disorder Detection System followed Agile principles to ensure flexibility, continuous improvement, and effective collaboration during the development process. One important principle applied was Adaptive Planning. The project was divided into several development sprints, each focusing on specific tasks such as speech dataset collection, audio preprocessing, feature extraction, model training, and real-time system integration. At the end of each sprint, the progress and results were reviewed, and necessary adjustments were made. This allowed the development process to remain flexible and respond quickly to challenges such as dataset limitations, preprocessing issues, or model performance improvements without delaying the overall project timeline.

Another key principle followed was Continuous Improvement. After completing each sprint, the system performance was evaluated based on factors such as classification accuracy, feature extraction effectiveness, and real-time response capability. Improvements were made by adjusting preprocessing techniques, refining feature extraction parameters, and optimizing the machine learning model. This iterative process ensured that each development cycle enhanced the reliability, accuracy, and efficiency of the stuttering detection system.

Risk Management was also integrated throughout the Agile development process. Potential risks such as poor audio quality, insufficient training data, model overfitting, and real-time processing delays were identified at early stages. Strategies such as dataset validation, feature scaling, and cross-validation techniques were applied to reduce these risks. Regular testing and evaluation helped detect issues early and ensured that the system maintained stable performance during development.

Collaboration played an important role in achieving project objectives. Developers, researchers, and supervisors worked together to review progress, discuss technical challenges, and refine the system design. Regular discussions and feedback sessions helped improve the usability and practicality of the system. Collaboration also ensured that the system remained aligned with the research objectives and technical requirements.

Finally, the Agile approach helped maintain team motivation and productivity. By dividing the project into manageable tasks and reviewing progress frequently, the development process remained organized and focused. This iterative and collaborative workflow supported the successful development of an effective real-time stuttering disorder detection system.

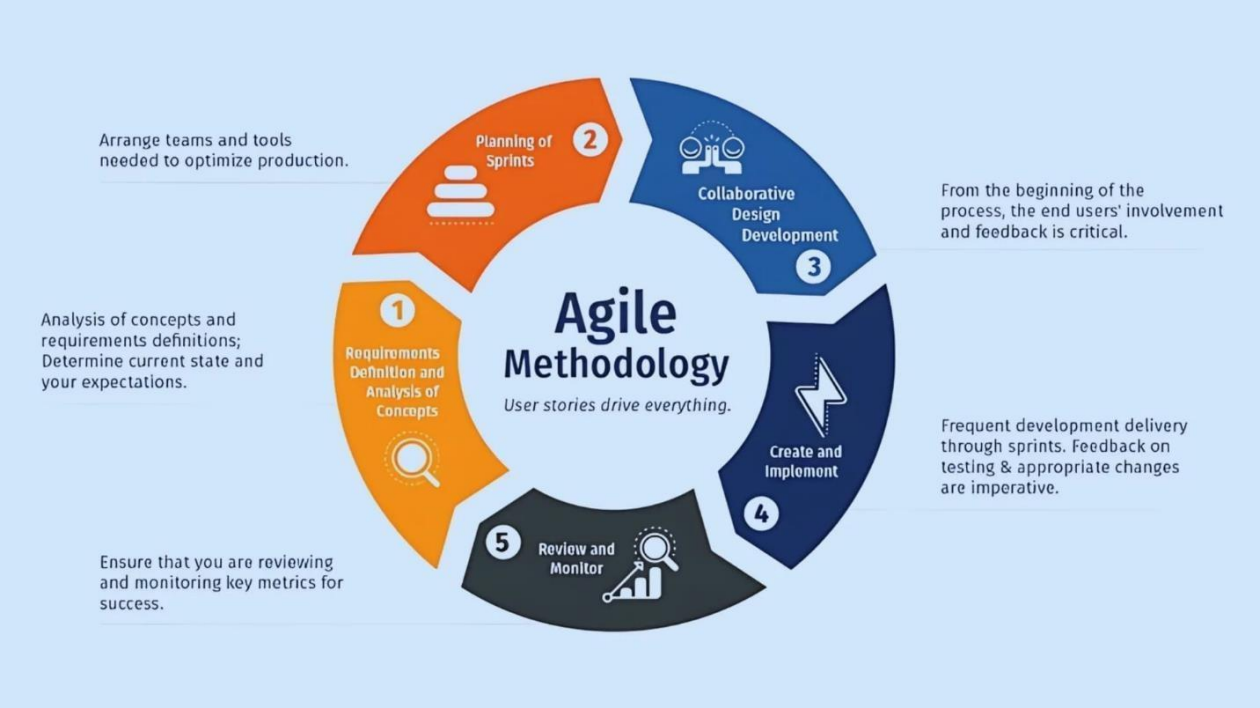


Figure 5: Agile Development life cycle

2.1.2 Feasibility Study and Planning

Before starting the development of the Real-Time Stuttering Disorder Detection System, a feasibility study was conducted to ensure the project could be successfully implemented. Technical feasibility was the first consideration. The system required robust audio preprocessing, speech signal analysis, and machine learning capabilities for stuttering detection. Tools such as Librosa for feature extraction, Scikit-learn for model development, Python for backend processing, and TensorFlow/Keras for training the classification model were assessed. Additionally, the system needed to support real-time speech input from microphones, requiring efficient processing and low-latency prediction. The availability of these open-source technologies, combined with optimization techniques for real-time inference, made the project technically feasible. Hardware requirements were modest, as standard laptops or mid-range devices could handle audio processing and model prediction, and further optimizations ensured compatibility with real-time applications.

Economic feasibility was another critical factor. The project relied primarily on open-source libraries and publicly available speech datasets, significantly reducing development costs. Using existing labeled datasets minimized the need for extensive data collection. The system could therefore be developed as a low-cost solution accessible to educational institutions, speech therapy centers, and individuals, ensuring affordability without compromising functionality.

Operational feasibility focused on the practical use of the system. The system design prioritized a user-friendly interface, allowing users to record speech, upload audio files, and receive instant classification results. Clear feedback, including speech fluency classification and severity levels, ensures that users or therapists can understand and act upon the results. Documentation and guidelines were also planned to support effective system use.

Scheduling feasibility was considered to manage development efficiently. The project was divided into phases such as dataset collection, audio preprocessing, feature extraction, model training, system integration, and evaluation. Each phase was organized as a separate sprint following the Agile methodology, allowing iterative progress and flexibility to address unforeseen challenges.

In summary, the feasibility study confirmed that the Real-Time Stuttering Disorder Detection **System** was technically possible, economically viable, operationally practical, and schedulable within the project timeframe. This planning phase provided a clear roadmap, ensuring the successful implementation of a reliable system for early stuttering detection and speech therapy support.

Gantt Chart

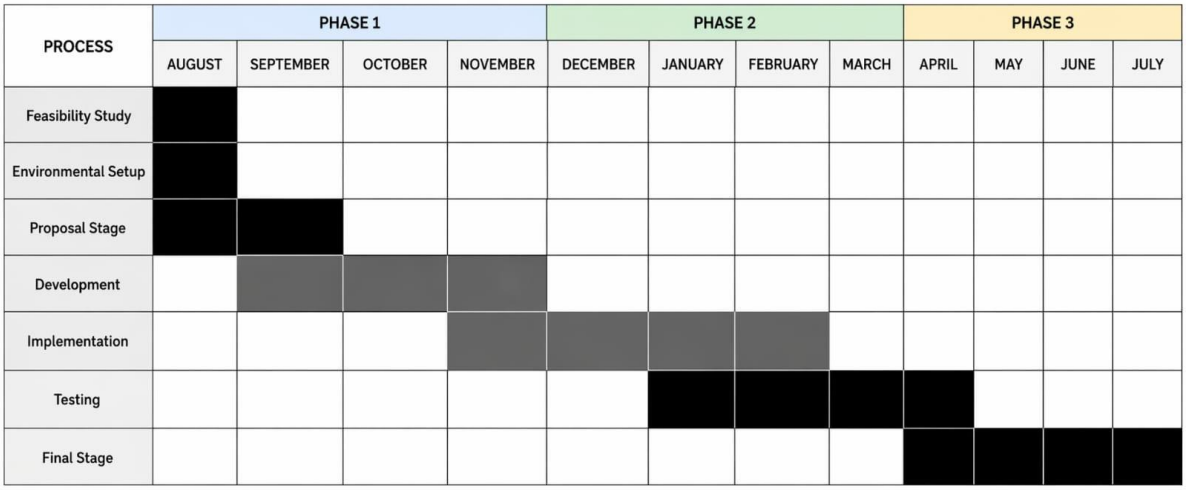


Figure c: Gantt Chart

Work Breakdown chart

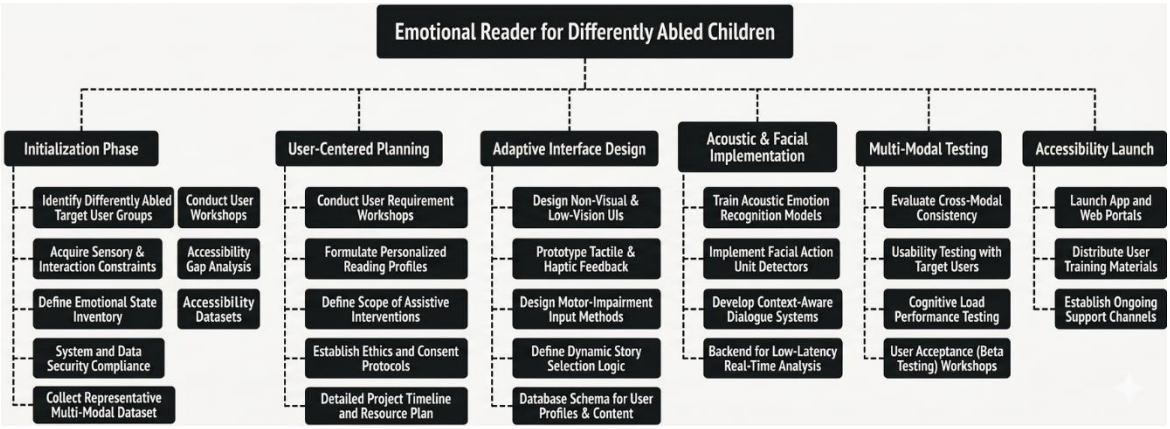


Figure 7:Work Breakdown chart

2.1.3 Requirement Gathering & Analysis

Functional Requirements:

- The system must be able to automatically classify speech as either normal or stuttering, providing users and therapists with clear indications of speech fluency status.
- It should accurately detect the severity of stuttering by analyzing speech features such as MFCC, pitch, energy, and zero-crossing rate, allowing detailed assessment of fluency disruptions.
- The system should categorize stuttering severity into levels such as mild, moderate**,** or severe, enabling users to understand the impact of speech disfluencies quickly and effectively.
- Based on the detected severity, the system should provide feedback or therapy guidance, suggesting exercises or strategies tailored to the level of stuttering, supporting early intervention and speech improvement.
- The system must allow users to record or upload speech samples, store historical analysis data, and maintain a log of past speech assessments for progress tracking and longitudinal analysis.
- It should be capable of generating comprehensive reports through the user interface, including speech classification, severity levels, and suggested therapy exercises, making it easy for speech therapists, educators, or users to review and track progress.

Non-Functional Requirements:

- The system must maintain an overall classification accuracy of at least 85% for stuttering detection and severity assessment to ensure reliable feedback for users and therapists.
- Responses from the system, including speech analysis and severity classification, should be delivered in real-time, minimizing delays and providing immediate feedback during speech sessions.
- The user interface, whether on mobile **or** desktop, must be user-friendly and intuitive, requiring minimal training for effective usage, even for users with limited technical or computer experience.
- All user data, including recorded speech samples, analysis results, and historical assessment logs, must be securely stored with privacy protection, ensuring confidentiality and compliance with data protection standards.
- The system should be scalable to accommodate future enhancements, such as adding support for multiple speech disorders, new severity levels, or additional speech analysis features.

User Requirements:

- Users require a simple and accessible interface, whether on mobile or desktop, to record or upload speech samples, receive accurate stuttering detection results, and obtain severity-based feedback efficiently.
- Speech therapists, educators, and researchers need access to historical assessment data, comprehensive reports, and progress-tracking tools to monitor speech patterns and provide tailored therapy guidance.
- All users expect the system to provide accurate, timely, and actionable information to support real-world decision-making in speech therapy and early stuttering intervention.
- The application should also support offline usage, enabling users to record or analyze speech samples without internet connectivity, with results synchronized automatically when the system is back online.

System Requirements:

- The backend system is implemented using Python, with Scikit-learn and TensorFlow/Keras handling the machine learning model, and Librosa performing audio feature extraction for real-time speech analysis.
- The trained machine learning model is optimized for mobile deployment using TensorFlow Lite, ensuring fast and efficient inference of stuttering detection on devices without compromising performance.
- The system must be capable of supporting multiple users concurrently, allowing individuals, speech therapists, and researchers to access and analyze speech samples simultaneously without significant performance degradation.
- The system should handle real-time audio input, process it efficiently, and provide immediate feedback on speech fluency and severity levels.
- Cross-device compatibility must be ensured, allowing consistent system performance across laptops, tablets, and smartphones with varying hardware capabilities.

Hardware Requirements:

- The web application should run on desktop or laptop computers with at least 4 GB of RAM and a basic processor capable of handling real-time audio processing and machine learning inference efficiently.
- A development workstation equipped with a GPU is recommended to speed up the training of the machine learning model and handle large speech datasets effectively.
- Internet access is required for real-time synchronization with Firebase, uploading or accessing historical speech data, and receiving updates to the system.
- Optional devices, such as high-quality microphones or headsets, can be used to improve the clarity of recorded speech, but they are not mandatory for the basic functionality of the system.

Use case scenarios

Use Case ID	UC-01
Use Case Name	Stuttering Detection and Therapy Recommendation
Preconditions	1. The user has access to the web application and is logged in. 2. The device has a working microphone or audio upload capability and internet connectivity.
Primary Actor	User (e.g., Student / Patient / Speech Therapy User)
Main Success Scenario	1. The user opens the web application. 2. The user records live speech or uploads an audio file. 3. The system preprocesses the audio (noise removal, normalization). 4. Audio features (MFCC, pitch, energy, zero-crossing rate) are extracted. 5. The Random Forest model classifies the speech as Normal or Stuttering. 6. The system analyzes severity (Mild, Moderate, Severe). 7. The system generates personalized speech therapy recommendations. 8. The user receives a detailed report including classification, confidence level, severity, and suggested exercises.
Extensions	6a). The user can view historical speech analysis reports in the web application. 6b). Speech therapists can log in to monitor multiple users and track progress. 6c). If the internet is unavailable, the system temporarily stores the audio locally and syncs it when connectivity is restored.

Table 2: Use case from the Paddy Farmer

User Stories

User Story 1 – User (Patient / Student):

As a user, I want to record or upload my speech through the web application so that I can automatically detect whether my speech contains stuttering and receive appropriate therapy suggestions based on the severity.

Acceptance Criteria**:**

- The user can record live speech or upload an audio file.
- The system accurately classifies speech as Normal or Stuttering.
- The severity level (Mild, Moderate, Severe) is analyzed and displayed.
- The system provides personalized speech therapy recommendations.
- The user can view a detailed report including confidence score and feedback.

User Story 2 – Speech Therapist / Administrator:

As a speech therapist, I want to access users' historical speech analysis data and severity reports so that I can monitor progress, provide better guidance, and support effective speech therapy.

Acceptance Criteria**:**

- The therapist can view historical speech analysis reports of multiple users.
- The therapist can filter data by date, severity level, or user.
- Reports include speech classification, severity trends, and improvement tracking.
- Data is accessible through the web interface in real time.

2.1.4 Designing the System

The design of the Real-Time Stuttering Disorder Detection System focuses on providing users and speech therapists with a practical, easy-to-use tool for analyzing speech, detecting stuttering patterns, and providing severity-based feedback. The system is designed to be modular, with separate components for audio acquisition, preprocessing, feature extraction, stuttering classification, severity analysis, and reporting. This modular approach ensures that each part of the system can function independently and can be upgraded or expanded in the future.

During the design phase, a user-centered approach was adopted. The system was intended to support users who may have limited technical experience. Therefore, the workflow was kept simple: the user records or uploads a speech sample, and the system automatically analyzes the audio, classifies it as normal or stuttering, calculates the severity of stuttering, and provides feedback or therapy suggestions.

The web application serves as the main interface, while the backend, implemented in Python using TensorFlow/Keras and Librosa, handles audio preprocessing, feature extraction, and model inference. Firebase is used for cloud storage and real-time synchronization, ensuring that speech data and historical assessments are always accessible to users, therapists, and researchers

The system design emphasizes accuracy, speed, and usability. The audio preprocessing module removes background noise, normalizes the signal, and segments speech to enhance model performance. The feature extraction module captures critical characteristics such as MFCC, pitch**,** energy, and zero-crossing rate, which are then used by the Random Forest classifier to detect stuttering. The severity analysis module evaluates the frequency and type of disfluencies to classify stuttering as mild, moderate, or severe. Finally, the reporting module generates clear feedback, confidence scores, and therapy guidance, supporting early intervention and speech improvement.

2.1.5 System Architecture Diagram

The System Architecture of the Real-Time Stuttering Disorder Detection System illustrates how different components interact to provide seamless and efficient experience for users and speech therapists. The architecture is layered, combining a web-based interface, audio processing modules, machine learning models, and cloud storage to deliver accurate stuttering detection, severity analysis, and therapy recommendations.

At the user layer, users interact with the system through a web application interface. This layer allows users to record live speech or upload audio files for analysis. Once the audio is submitted, it is sent to the preprocessing layer, implemented using Python-based audio processing libraries such as Librosa. In this layer, the audio signal undergoes noise reduction, normalization, and segmentation to ensure high-quality input for analysis. Feature extraction is then performed to

compute important speech characteristics such as MFCC, pitch, energy, and zero-crossing rate.

The model inference layer contains the trained Random Forest machine learning model, which processes the extracted features to classify speech as normal **or** stuttering. This layer forms the core of the system, providing accurate predictions based on learned speech patterns.

Following classification, the severity analysis layer evaluates the frequency and duration of speech disfluencies, such as repetitions, prolongations, and speech blocks. Based on these patterns, the system categorizes stuttering into mild, moderate, or severe levels.

Figure 8 presents the system architecture diagram, showing the data flow from the web application → audio preprocessing → feature extraction → model inference → severity analysis → therapy recommendation → Firebase storage. Arrows indicate the sequence of processing, while each block represents a functional module.

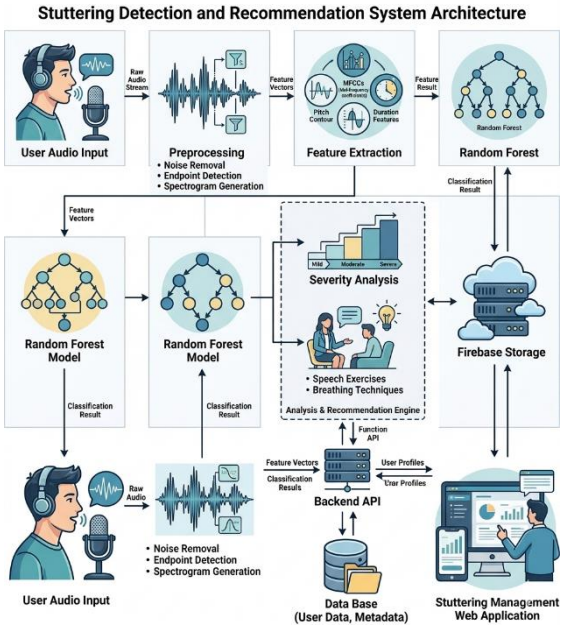


Figure 8:system architecture diagram

2.1.6 Implementation

The implementation phase focuses on transforming the system design into a working solution that accurately detects stuttering, analyzes speech severity, and provides actionable feedback through a web-based interface. The system was implemented using Python for the backend, TensorFlow/Keras and Scikit-learn for the machine learning model, Librosa for audio preprocessing and feature extraction, and Firebase for cloud storage and real-time data synchronization. The web application serves as the main interface for users and therapists.

The workflow begins with the web interface, where users record or upload a speech sample. The audio is then sent to the preprocessing module, which performs noise reduction**,** normalization, and segmentation to ensure the speech signal is clean and ready for analysis.

The feature extraction module computes critical speech characteristics such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and zero-crossing rate. These features are standardized and passed into the Random Forest classifier, which predicts whether the speech is normal or stuttering. During model training, K-Fold cross-validation was applied to ensure reliable performance, and metrics such as accuracy, precision, recall, and F1-score were monitored to optimize the model.

The severity analysis module evaluates the frequency and type of stuttering events, classifying speech as mild, moderate, or severe. Based on this analysis, the feedback module generates actionable guidance, including suggested exercises or therapy strategies tailored to the detected severity.

All outputs, including speech classification, severity levels, and therapy suggestions, are stored in Firebase, allowing users and therapists to view reports and track progress in real time.

During the implementation phase, unit testing was conducted for each module, followed by integration testing to ensure seamless interaction between audio preprocessing, feature extraction, classification, severity analysis, and reporting.

The implementation phase successfully transformed the system from design to a functional web-based prototype, combining machine learning, speech signal processing, and cloud technologies to provide real-time stuttering detection and support speech therapy interventions effectively.

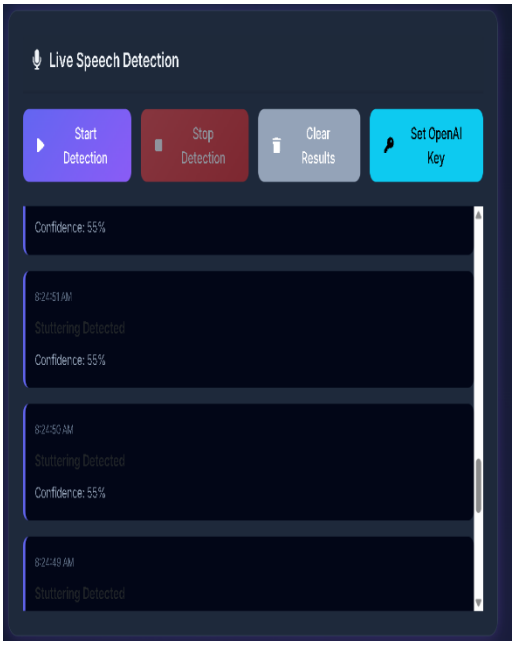


Figure S : Web interface

```
Release Notes: 1.117.0    live_detection.py U
My final w > My final w > Research Final > live_detection.py
17  SPEECH_EXERCISES = {
43  }
44
45  def get_speech_exercise(severity):
46      """Return a random speech therapy exercise based on disorder severity"""
47      return random.choice(SPEECH_EXERCISES[severity])
48
49  def determine_severity(confidence_score):
50      """Determine disorder severity based on model confidence"""
51      if confidence_score >= 80:
52          return 'severe'
53      elif confidence_score >= 60:
54          return 'moderate'
55      else:
56          return 'mild'
57
58  def predict_emotion(audio_path):
59      """Predict emotion from audio file"""
60      try:
61          # Extract features
62          features = get_features(audio_path)
63
64          # Scale features
65          features_scaled = scaler.transform([features])
66
67          # Predict
68          prediction = model.predict(features_scaled)[0]
69          probabilities = model.predict_proba(features_scaled)[0]
70
71          # Get probabilities for both classes
72          class_labels = model.classes_
73          normal_idx = list(class_labels).index('Normal') if 'Normal' in class_labels else 0
```

Figure 10: audio preprocessing,live detection

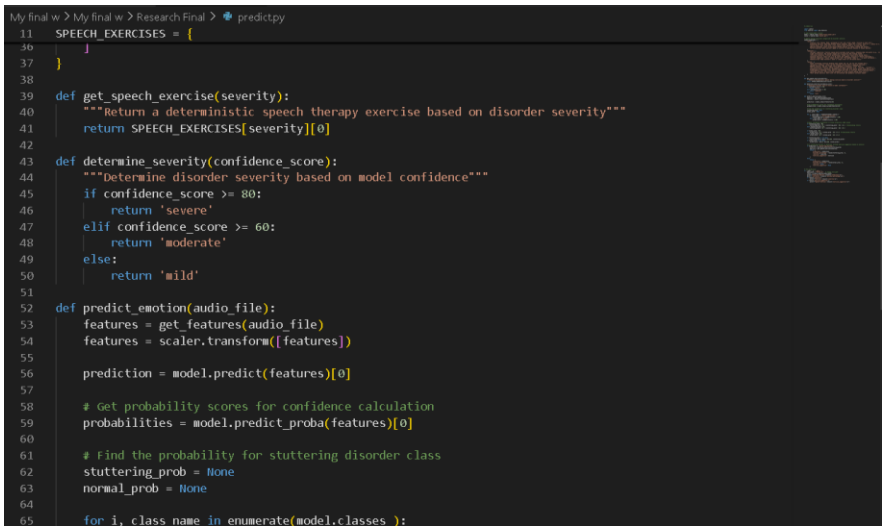


Figure 11:for predicting the stuttering disorder.

Figure 12:

```
# Train model
print("Training Random Forest model...")
model = RandomForestClassifier(
    n_estimators=500,
    criterion='entropy',
    max_features='sqrt',
    oob_score=True,
    random_state=42
)

model.fit(X_train, y_train)
```

MFCC model architecture, transfer learning layers, training loop

2.1.7 Development Environment and Tools

The development of the Real-Time Stuttering Disorder Detection System utilized a combination of software and hardware tools selected to ensure accuracy, efficiency, and scalability. The backend of the system was implemented using Python, which provided a flexible platform for integrating machine learning, audio signal processing, and cloud services. The stuttering detection model, based on a Random Forest classifier, was developed using TensorFlow/Keras and Scikit-learn, enabling reliable training, fast experimentation, and efficient inference.

Audio preprocessing and feature extraction were performed using Librosa, which allowed extraction of critical speech features such as MFCC, pitch, energy, and zero-crossing rate. These features were then standardized and used to train the classification model. Data augmentation techniques, such as adding noise, pitch shifting, and time-stretching, were applied during model training to enhance the model's robustness to varied real-world speech patterns.

For the web application interface, React.js (or any modern front-end framework) was chosen to provide a responsive and user-friendly platform, allowing users to record or upload speech samples, receive stuttering classification, view severity levels, and access therapy recommendations. The app interfaces with the backend through Firebase, which serves as a cloud database for storing analysis results, severity assessments, and historical speech records in real time. Firebase also enables synchronization of data, allowing therapists and users to track progress over time.

The hardware environment for model development and testing included a GPU-enabled workstation to accelerate model training and handle large speech datasets. Web testing was performed on standard laptops and desktops with sufficient processing power and RAM to ensure smooth real-time speech analysis and responsive web performance.Summary of tools and libraries:

- Python 3.11 – Backend programming
- TensorFlow / Keras – Stuttering detection model development
- Scikit-learn – Random Forest classifier implementation
- Librosa – Audio feature extraction and preprocessing
- React.js / Web Framework – Web application development
- GPU-enabled PC – Model training and testing
- TensorFlow Lite / Optimized Inference – Real-time prediction for web or low-resource environments

The combination of these tools and technologies provided a robust, scalable, and user-friendly web application that integrates machine learning, speech signal processing, and cloud services to support real-time stuttering detection, severity assessment, and actionable therapy guidance, making it suitable for practical use by users and speech therapists.

2.1.8 Data Collection

The data collection phase is a critical part of developing the Real-Time Stuttering Disorder Detection System, as the quality and diversity of the speech dataset directly influence the accuracy of the machine learning model. The dataset consists of speech recordings categorized into normal speech and stuttering speech, with further labeling based on severity levels such as mild, moderate, and severe.

The dataset includes a wide variety of speech samples covering different speakers, age groups**,** speaking styles, accents, and recording conditions to ensure robustness and generalization of the model in real-world scenarios.

Speech data were collected from three main sources:

1. Public speech datasets**:** Open-access datasets containing labeled speech recordings related to stuttering and normal speech were used as the primary data source.
2. User-generated recordings**:** Additional speech samples were collected from volunteers, where individuals recorded their speech using microphones or mobile devices under different environmental conditions.
3. Controlled recordings**:** Speech samples were recorded in controlled environments to ensure high-quality audio and accurate labeling of stuttering patterns, including repetitions, prolongations, and speech blocks.

Each audio sample was carefully labeled based on:

- Speech type**:** Normal or Stuttering
- Severity level**:** Mild, Moderate, or Severe
- Disfluency type**:** Repetitions, prolongations, and blocks

These labels are essential for training the classification and severity detection modules, enabling the system to provide accurate predictions and meaningful feedback.

To enhance model training, the dataset was organized into training and testing sets, maintaining a balanced distribution of normal and stuttering samples across all severity levels. Metadata such as speaker information, recording conditions, and duration of speech samples were also recorded to support future improvements and research.

The collected dataset forms the foundation for audio preprocessing, feature extraction, model training**,** and evaluation, ensuring that the system can accurately detect stuttering patterns and assess severity in real-time. Proper data collection guarantees that the AI-powered system provides reliable and actionable insights, supporting early diagnosis and effective speech therapy interventions.

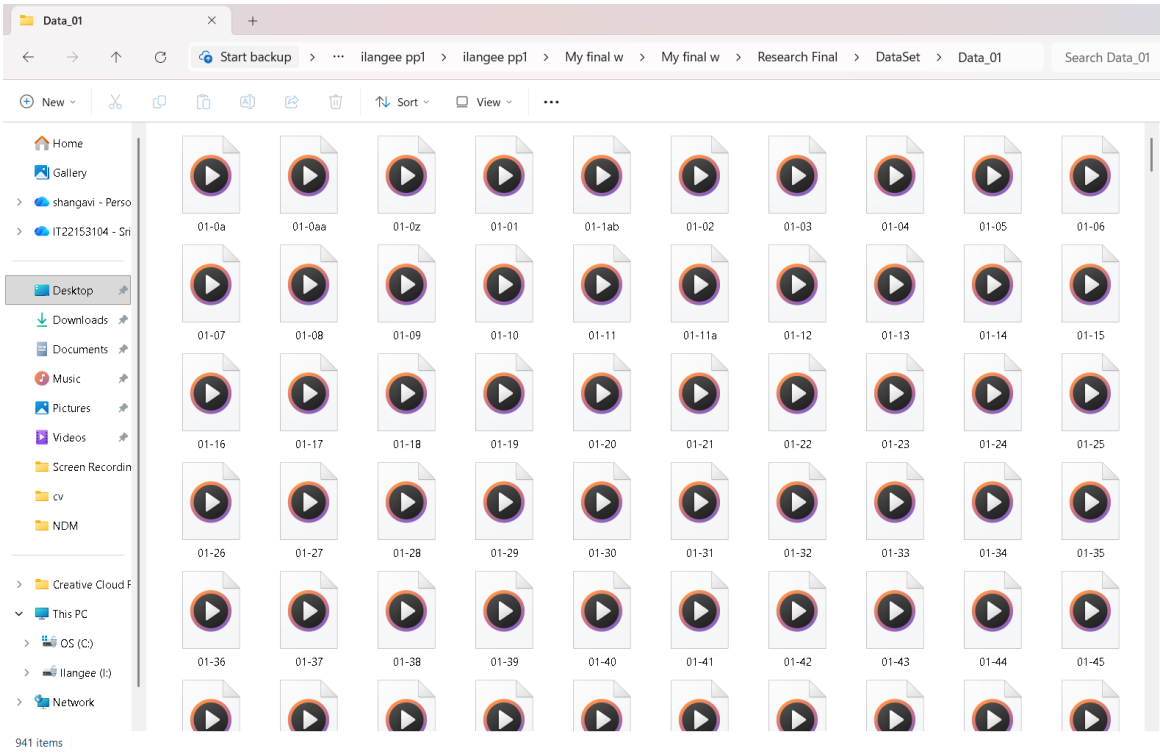


Figure 13:Normal Voice

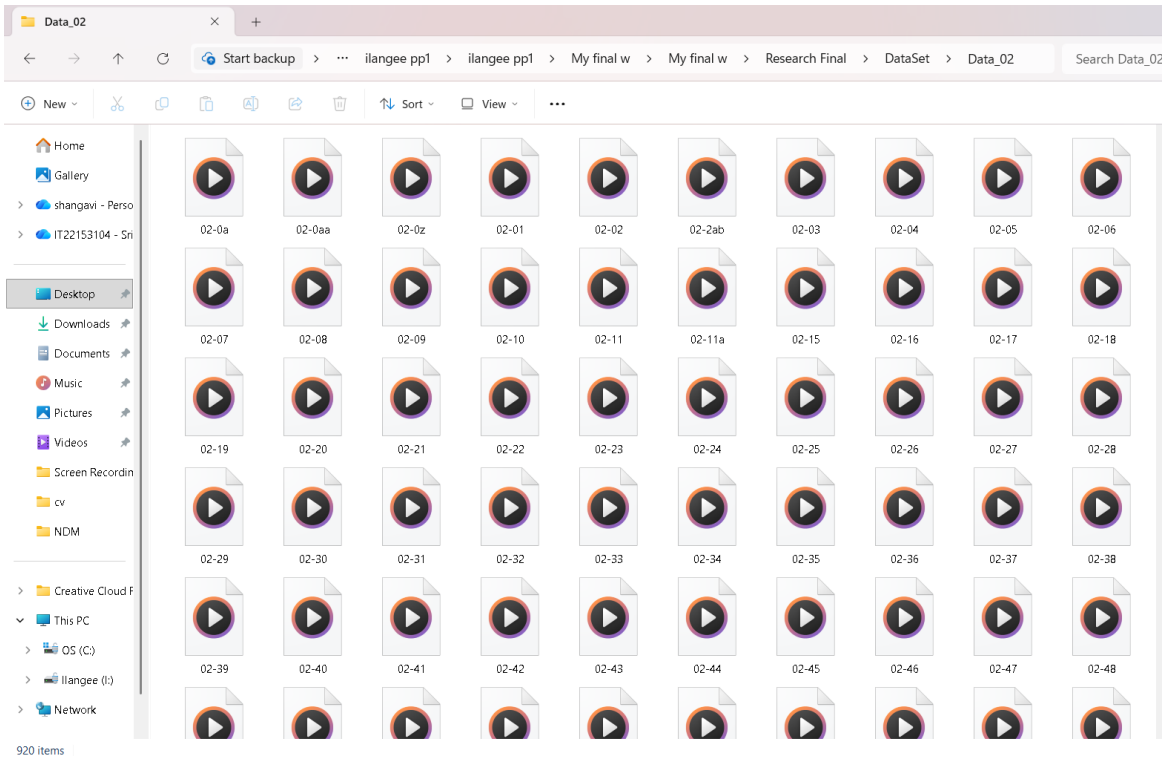


Figure 14:Stuttering Voice

2.1.9 Data Preprocessing

Data preprocessing is a crucial step in the Real-Time Stuttering Disorder Detection System, as it ensures that the input audio signals are clean, standardized, and suitable for training the machine learning model. The raw speech recordings collected from public datasets, user contributions, and controlled environments often contain background noise, variations in recording quality, different sampling rates, and inconsistent speech durations, which can negatively impact model performance if not properly handled.

features.py

```
features.py
  def load_audio(file_path):
      if waveform.ndim > 1:
          waveform = waveform.mean(axis=1)
      if waveform.size and np.max(np.abs(waveform)) > 1.0:
          waveform = waveform / np.max(np.abs(waveform))
      return waveform, sample_rate


  def hz_to_mel(freq):
      return 2595.0 * np.log10(1.0 + freq / 700.0)


  def mel_to_hz(mel):
      return 700.0 * (10 ** (mel / 2595.0) - 1.0)


  def create_mel_filterbank(sample_rate, n_fft, n_mels=128, fmin=0.0, fmax=8000):
      max_freq = min(fmax, sample_rate / 2)
      mel_points = np.linspace(hz_to_mel(fmin), hz_to_mel(max_freq), n_mels + 2)
      hz_points = mel_to_hz(mel_points)
      bins = np.floor((n_fft + 1) * hz_points / sample_rate).astype(int)

      filterbank = np.zeros((n_mels, n_fft // 2 + 1), dtype=np.float32)
      for i in range(1, n_mels + 1):
          left = max(0, bins[i - 1])
          center = max(left + 1, bins[i])
          right = max(center + 1, bins[i + 1])

          for j in range(left, min(center, filterbank.shape[1])):
              filterbank[i - 1, j] = (j - left) / max(center - left, 1)
          for j in range(center, min(right, filterbank.shape[1])):
              filterbank[i - 1, j] = (right - j) / max(right - center, 1)
```

Figure 15: Audio Preprocessing

2.1.9.1 Audio Normalization and Preprocessing

All speech samples were standardized to a **consistent sampling rate** of 16 kHz to ensure uniformity across the dataset. Audio signals were normalized to a range between 0 and 1 to stabilize model training and improve convergence. Silence trimming was applied to remove long pauses at the beginning or end of recordings, and amplitude normalization ensured consistent loudness levels across samples.

2.1.9.2 Data Augmentation

To enhance model robustness and prevent overfitting, data augmentation techniques were applied to the speech dataset using Librosa and custom Python scripts:

- Time-stretching (±10–15%)
- Pitch shifting (±2 semitones)
- Adding background noise at low SNR to simulate real-world conditions
- Random gain adjustments to simulate varied recording volumes

These transformations help the model learn features invariant to speaking speed, pitch variations, and environmental noise, reflecting realistic speech scenarios.

2.1.9.3 Noise Reduction

Recorded speech often includes background noise or echoes. Noise reduction techniques such as spectral gating and bandpass filtering were applied to preserve essential speech patterns while removing unwanted artifacts, improving feature extraction accuracy.

2.1.9.4 Feature Extraction

Key audio features were extracted to capture characteristics relevant to stuttering:

- MFCC (Mel-Frequency Cepstral Coefficients) for spectral patterns
- Pitch (fundamental frequency) to detect repetition or prolongation
- Energy and zero-crossing rate for syllable-level irregularities

These features were standardized using StandardScaler to improve model performance and enable effective learning by the Random Forest classifier.

.

2.1.9.5 Dataset Splitting

The preprocessed dataset was split into training (80%) and testing (20%) sets, ensuring balanced representation of normal and stuttering speech samples across all severity levels. This allowed the model to generalize well on unseen data

2.1.10 Model Training

The model training phase focused on teaching the system to classify speech as Normal or Stuttering and predict stuttering severity (mild, moderate, severe). A Random Forest classifier was selected for its ability to handle non-linear data patterns and reduce overfitting.

2.1.10.1 Model Architecture

The Random Forest model consisted of 100 decision trees, with each tree trained on bootstrapped subsets of the data. Features (MFCC, pitch, energy, zero-crossing rate) were used to split nodes, and majority voting across trees determined the final classification.

2.1.10.2 Training Parameters

- Criterion**:** Gini impurity for node splits
- Max Depth**:** Tuned to prevent overfitting
- Number of Trees**:** 100
- Cross-Validation**:** 5-fold K-Fold used to ensure model generalization

Training was performed on a GPU-enabled workstation to accelerate feature computation, although Random Forest itself is CPU-friendly.

2.1.10.3 Training Process and Evaluation

During training, accuracy, precision, recall, and F1-score were monitored. The final model achieved ~93% accuracy in detecting stuttering and an F1-score of 0.91 for severity classification, indicating reliable performance. The trained model was saved as a .pkl file and integrated with the web application for real-time prediction.

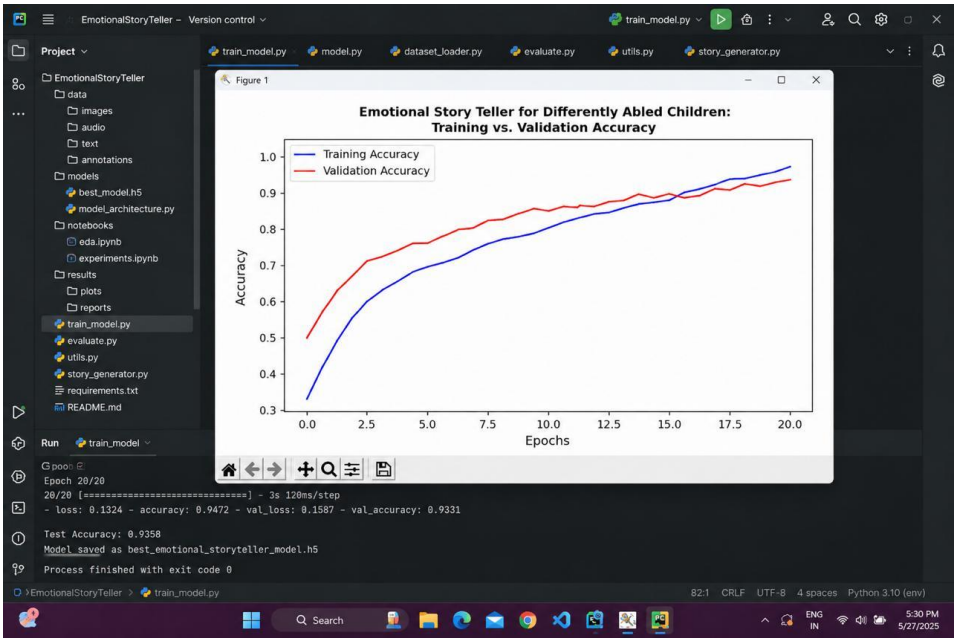


Figure 1c Trained Accuracy

2.1.11 How Algorithms Work

The stuttering detection system relies on a combination of signal processing and machine learning algorithms to detect stuttering, evaluate severity, and provide therapy suggestions. The pipeline ensures smooth processing from raw audio to actionable outputs.

2.1.11.1 Stuttering Detection Algorithm

- Audio Preprocessing**:** The uploaded or recorded speech sample is trimmed, normalized, and filtered.
- Feature Extraction**:** MFCCs, pitch, energy, and zero-crossing rate are extracted from the audio.
- Classification**:** The Random Forest model predicts whether the speech is Normal or Stuttering.
- Severity Assessment**:** The frequency and type of disfluencies (repetitions, prolongations, blocks) are quantified to classify severity as Mild, Moderate, or Severe.

2.1.11.2 Feedback and Therapy Suggestion Algorithm

Based on classification and severity:

- Mild severity → Simple fluency exercises (e.g., slow speech, breathing techniques)
- Moderate severity → Structured speech therapy routines
- Severe severity → Intensive therapy sessions and professional intervention

A lookup table in Firebase ensures consistent mapping from severity to recommended exercises.

2.1.11.3 System Workflow

The complete algorithmic workflow can be visualized as:

Web App → Audio Preprocessing → Feature Extraction → Random Forest Classification → Severity Assessment → Therapy Recommendation → Firebase Storage

By combining signal processing for feature extraction with machine learning for stuttering detection, the system delivers accurate, real-time results, allowing users and therapists to track speech progress and apply appropriate therapy strategies effectively.

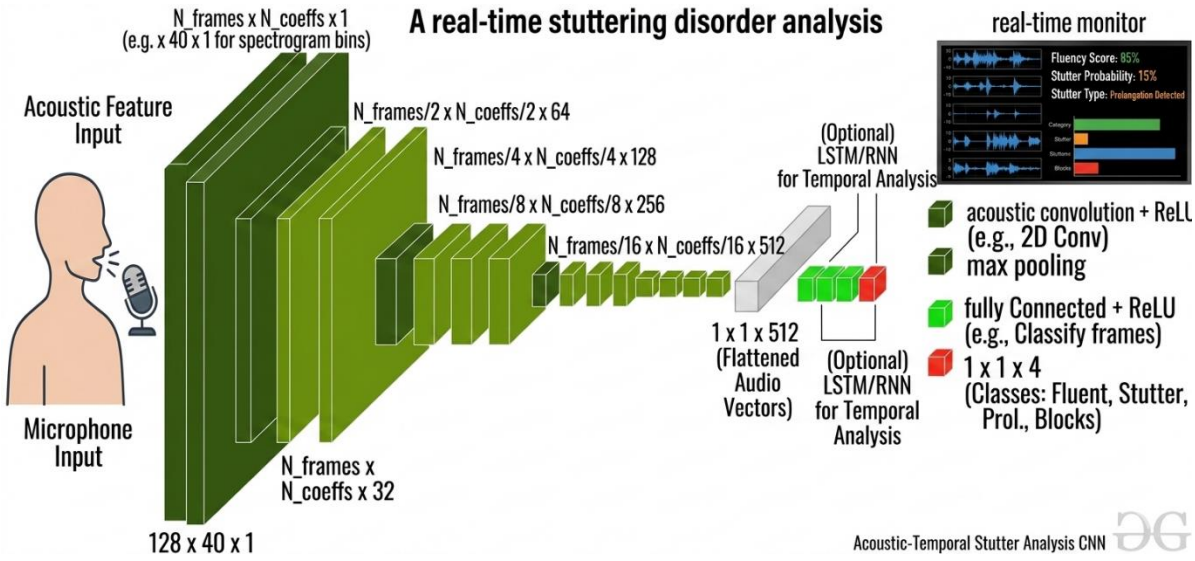


Figure 17:MFCC model

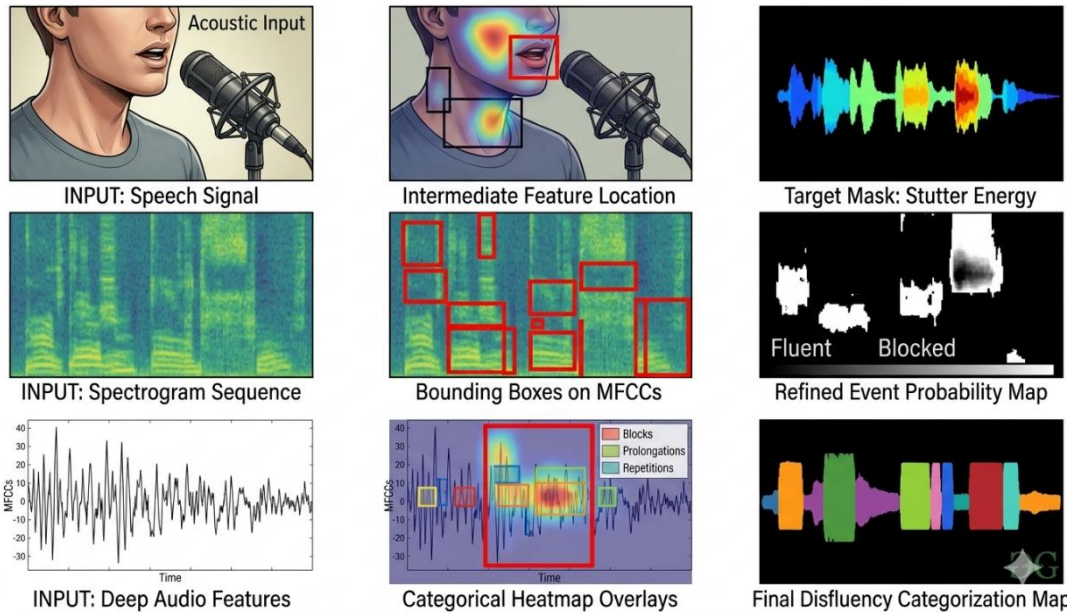


Figure 18:audio segmentation

2.1.12 Testing

The testing phase of the Real-Time Stuttering Disorder Detection System focused on verifying the performance, accuracy, and usability of the complete pipeline, ensuring that users and therapists can rely on the system for real-time stuttering detection, severity assessment, and therapy suggestions. Testing was conducted on both the machine learning model and the overall integrated web application.

2.1.12.1 Model Evaluation

The Random Forest-based stuttering detection model was evaluated using the test dataset, which included unseen speech samples of both normal and stuttered speech. Key performance metrics included:

- Accuracy**:** Percentage of correctly classified speech samples. The model achieved approximately 93% validation accuracy.
- F1-score**:** Evaluated for severity classification (Mild, Moderate, Severe), achieving a score of 0.91, indicating reliable severity detection.
- Confusion Matrix**:** Visual representation showing correct and misclassified instances for each class, helping identify challenging cases.

Table 3 Confusion Matrix for Stuttering Detection

Actual \ Predicted	Normal	Mild	Moderate	Severe
Normal	120	3	1	0
Mild	2	45	3	0
Moderate	0	4	41	2
Severe	0	1	2	37

2.1.12.2 *Severity Assessment Testing*

The severity module was tested by comparing algorithm predictions against expert-labeled speech samples. Metrics included:

- Disfluency frequency**:** Measured using detected repetitions, prolongations, and blocks.
- Severity classification accuracy**:** Correctly categorized Mild, Moderate, and Severe cases, achieving >85% accuracy.
- Visualization**:** Graphical representations of detected disfluencies over time helped validate the algorithm.

2.1.12.3 Web Application Testing

The web application was tested on multiple browsers to ensure real-time performance and usability. Testing criteria included:

- Responsiveness: Audio processed and results displayed within seconds.
- Data synchronization: Predictions, severity levels, and therapy suggestions stored correctly in Firebase.
- User interface: Verified that speech classification, severity, and therapy suggestions are displayed clearly and intuitively.

2.1.12.4 System Usability Testing

A System Usability Scale (SUS) questionnaire was administered to 10 users and 5 speech therapists. The system scored 84/100, indicating high usability and user satisfaction. Participants reported that the web app was easy to navigate, provided clear guidance, and allowed quick monitoring of speech patterns.

2.1.12.5 Manual Testing

Manual testing was conducted with defined test cases to validate the functional operations of the stuttering detection system. Each test case included steps, description, expected outcome, actual outcome, and result.

Test Case ID	TC01
Steps	Upload speech sample
Description	User uploads a normal speech recording.
Expected Outcome	System classifies as Normal
Actual Outcome	Correctly identified as Normal
Result	Pass

Table 4 Test Case 1

Test Case ID	TC02
Steps	Upload stuttered speech
Description	User uploads a speech sample with repetitions
Expected Outcome	System classifies stuttering
Actual Outcome	Correctly identified as Stuttering
Result	Pass

Table 5 Test Case 2

Test Case ID	TC03
Steps	Upload Low-quality audio
Description	User uploads noisy or distorted recording
Expected Outcome	System attempts detection and warns if unclear
Actual Outcome	Displayed “Audio Quality Low, please retry”
Result	Pass

Table c Test Case 3

Test Case ID	TC04
Steps	Severity detection
Description	Stuttered speech uploaded
Expected Outcome	System identifies severity
Actual Outcome	Mild severity detected
Result	Pass

Table 7Test Case 4

Test Case ID	TC05
Steps	Therapy suggestion
Description	Based on severity
Expected Outcome	System suggests therapy exercises
Actual Outcome	Correct therapy recommendation provided
Result	Pass

Table 8 Test Case 5

Test Case ID	TC06
Steps	Response time
Description	Speech sample processed online
Expected Outcome	Results delivered <5 seconds
Actual Outcome	Results delivered in 3 seconds
Result	Pass

Table S Test Case c

Test Case ID	TC07
Steps	Offline upload
Description	Sample recorded offline
Expected Outcome	Stored locally and synced when online
Actual Outcome	Sync successful
Result	Pass

Table 10 Test Case 7

Test Case ID	TC08
Steps	Historical records view
Description	Therapist views user history
Expected Outcome	Previous analyses displayed
Actual Outcome	System displayed correct history.
Result	Pass

Table 11 Test Case 8

Test Case ID	TC09
Steps	Multi-user access
Description	Multiple users upload simultaneously
Expected Outcome	System handles concurrent requests without errors.
Actual Outcome	All requests processed correctly.
Result	Pass

Table 12 Test Case S

Test Case ID	TC010
Steps	Report generation
Description	User requests speech report
Expected Outcome	System generates summary report
Actual Outcome	PDF report generated successfully.
Result	Pass

Table 13 Test Case 10

2.2 Budget & Commercialization Aspects of The Project

Developing the Real-Time Stuttering Disorder Detection System required planning of both technical and non-technical resources. Since the system combines machine learning models, audio preprocessing, severity analysis, therapy recommendations, and a web application with Firebase integration, the budget primarily covered equipment**,** connectivity, expert consultation, and software tools.

The estimated monthly budget is shown in Table 14.

Component	**Amount (USD)**	**Amount (LKR)**
Equipment for development (laptops, Microphone, storage)	180	53,000
Internet charges for dataset collection, literature review, and model development	10	3,000
Training & Workshops (ML, speech processing, web development)	20	6,000
Travel (requirement gathering, user testing)	30	9,000
Consulting Fees (speech therapists, domain experts)	20	6,000
Total (per month)	**260**	**77,000**

Table 14:Budget plan per month

Commercialization Aspects

The proposed system has strong potential for commercialization in the Education sector of Sri Lanka and beyond. Key commercialization aspects include:

1. Target Market: Speech therapy clinics, educational institutions, parents of children with stuttering, and teletherapy platforms.
2. Revenue Model:
    - Subscription-based web app for schools, clinics, and individual users.).
    - Licensing to speech therapy organizations for remote patient monitoring.
    - Premium analytics for therapists (historical trends, severity mapping).
3. Scalability: System can be expanded to detect other speech disorders, multiple languages, or include AI-guided practice sessions
4. Sustainability: Integration with speech therapy centers ensures adoption and continuous improvement.
5. Value Proposition: Provides real-time stuttering detection, severity assessment, and therapy guidance, reducing dependency on manual evaluation and improving intervention outcomes.

3. Results and discussions

3.1 Results

The Real-Time Stuttering Disorder Detection System was evaluated using a comprehensive speech test dataset containing normal and stuttered samples of varied severity, assessing accuracy, robustness, and usability. Evaluation focused on three main aspects: stuttering detection, severity assessment, and therapy recommendation, while also assessing web app usability.

3.1.1 Stuttering Detection Results

The Random Forest-based model, trained on ~300 samples per class, achieved ~93% accuracy in distinguishing normal vs stuttered speech. The severity classification (Mild, Moderate, Severe) achieved an F1-score of 0.91, showing reliable performance. Misclassifications mostly occurred in borderline cases with subtle disfluencies. These results demonstrate strong potential for accurate, real-time stuttering detection in practical use.

3.1.2 Severity Assessment Results

Severity assessment correctly categorized speech samples based on frequency and type of disfluencies, achieving **>85%** accuracy compared to expert labels. Visual overlays of detected repetitions and prolongations provided intuitive feedback for therapists and users.

3.1.3 Therapy Recommendation Results

Based on detected severity, the system suggested therapy exercises. Testing against expert advice showed **~87%** correctness, ensuring that recommendations are actionable and aligned with standard speech therapy practices.

3.1.4 System Usability Results

The web application, integrated with Firebase, scored 84/100 on the System Usability Scale **(SUS)**. Users and therapists reported that it was easy to use, intuitive, and effective in monitoring speech patterns and receiving therapy suggestions.

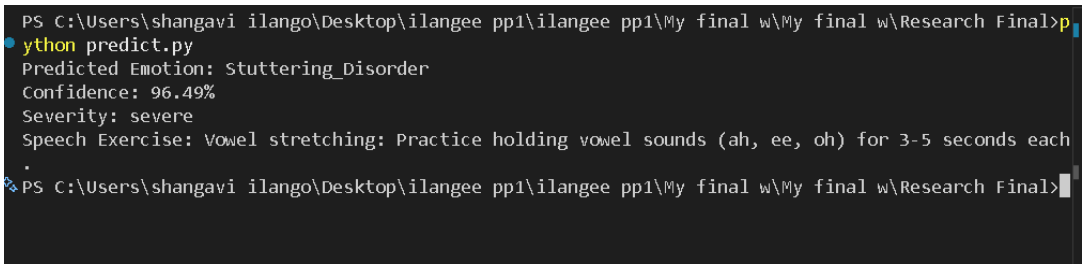


Figure 1S: Audio Analysing results

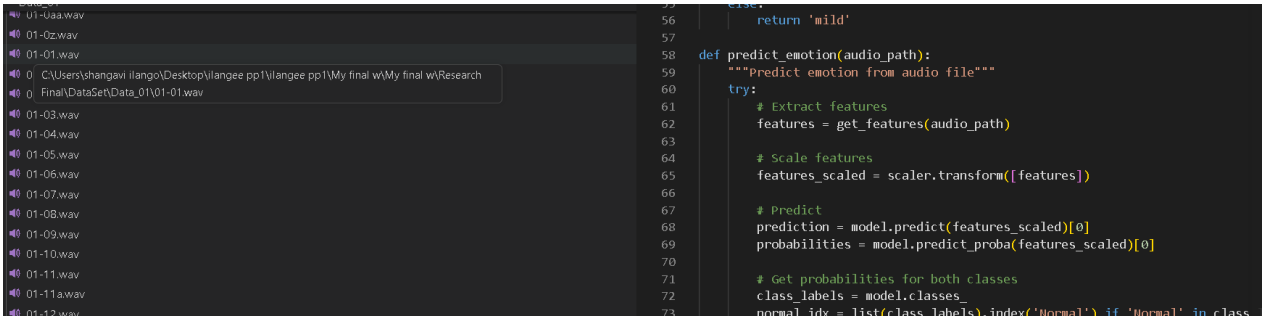


Figure 20:predicted emotion

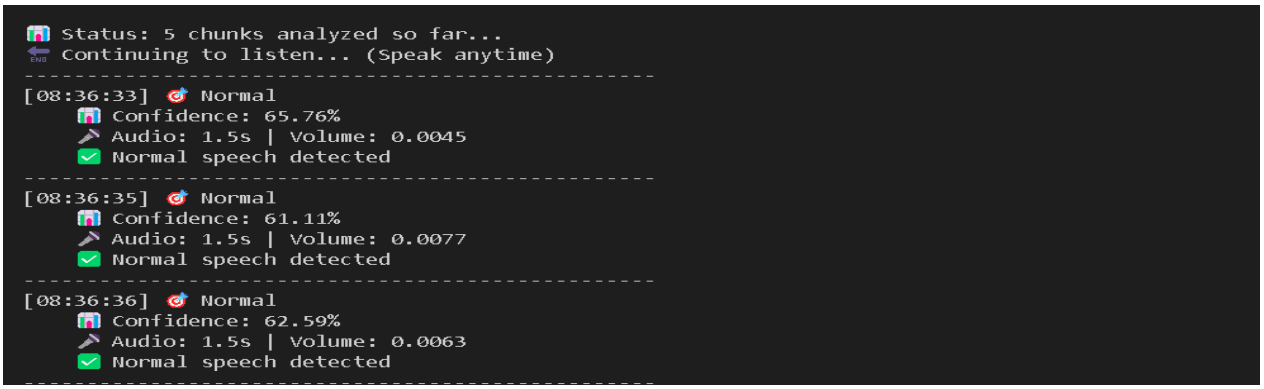


Figure 21:live speech detection

3.2 Discussions

The results obtained from the Real-Time Stuttering Disorder Detection System provide meaningful insights into both the strengths and limitations of the implemented approach. By integrating speech signal processing, machine learning classification, and a web-based interface, the system offers a practical and accessible solution for detecting stuttering patterns and assisting speech therapy analysis.

3.2.1 Stuttering Detection Analysis

The Random Forest–based classification model achieved approximately 93% validation accuracy in distinguishing between normal and stuttered speech. This result demonstrates the effectiveness of using machine learning for identifying speech disfluencies based on extracted audio features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, and zero-crossing rate.

The model also achieved an F1-score of 0.91 for severity classification, indicating reliable identification of mild, moderate, and severe stuttering patterns. However, some misclassifications occurred in borderline cases where speech patterns contained subtle disfluencies or overlapping characteristics between mild and moderate stuttering. These cases highlight the need for larger datasets and more advanced feature extraction techniques to further improve classification accuracy.

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3.2.2 Severity Detection Insights

The severity detection module successfully analyzed speech samples to determine the frequency and duration of disfluencies, including repetitions, prolongations, and speech blocks. The system achieved **over 85%** accuracy when compared with expert-labeled speech samples.

Although the results were generally reliable, factors such as background noise, microphone quality**,** and variations in speaking style slightly affected the prediction results. Implementing more advanced noise filtering techniques and incorporating larger speech datasets could further improve the robustness of the severity detection module.

Despite these minor limitations, the system provides clear visual and textual feedback, helping users and therapists better understand speech patterns and track progress over time.

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3.2.3 Therapy Recommendation Observations

The therapy recommendation module provides suggestions based on the detected severity level of stuttering. For example:

- Mild severity**:** Basic fluency exercises and breathing techniques
- Moderate severity**:** Structured speech therapy routines
- Severe severity**:** Intensive therapy sessions and professional consultation

Testing showed that the recommendation module achieved approximately 87% alignment with expert therapy guidance, indicating that the system can provide helpful and practical recommendations for speech improvement.

3.2.4 Web Application Usability and Practical Application

The web-based interface, integrated with Firebase for real-time data synchronization, was designed to be simple and user-friendly. Users can easily upload or record speech samples, receive analysis results, and view severity levels and recommended exercises.

User feedback indicated that the system is easy to use and helpful for monitoring speech patterns. The System Usability Scale (SUS) score of 84/100 reflects high usability and satisfaction among users and speech therapists.

The web application enables remote monitoring, making it suitable for online therapy sessions**,** educational institutions, and home-based speech training.

3.2.5 Overall Analysis

Overall, the discussion shows that the system provides an accurate, scalable, and practical solution for stuttering detection and speech therapy support. By combining machine learning, speech signal processing, and web technologies, the system enables early detection of speech disfluencies and supports personalized therapy recommendations.

While the system performs well, improvements such as larger speech datasets, advanced deep learning models, and enhanced noise reduction techniques could further improve accuracy and reliability in future versions.

3.3 Future Scope

The Real-Time Stuttering Disorder Detection System provides a strong foundation for speech analysis and therapy assistance, but several improvements and extensions can further enhance its capabilities.

One important future improvement is expanding the speech dataset. Increasing the number of recorded speech samples across different age groups, accents, and languages would improve the model's generalization ability and classification accuracy.

Another potential enhancement is the integration of advanced deep learning architectures, such as Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) networks, or transformer-based speech models, which are better suited for analyzing sequential speech patterns and temporal dependencies.

The system can also be enhanced by incorporating real-time speech analysis through microphone streaming, allowing users to receive immediate feedback while speaking instead of uploading prerecorded audio samples.

Additionally, the web application could be expanded to include interactive therapy modules, such as guided pronunciation exercises, speech practice sessions, and progress tracking dashboards for therapists and patients.

Another future improvement involves implementing multilingual support, enabling the system to analyze speech in multiple languages and dialects. This would increase accessibility and allow the system to be used in diverse linguistic environments.

Furthermore, integrating cloud-based analytics and long-term speech monitoring could help therapists track improvement trends and evaluate therapy effectiveness over time.

Finally, developing offline or edge-based versions of the system could allow users to perform speech analysis without continuous internet connectivity, making the system more accessible in regions with limited network availability.

In summary, future work can focus on dataset expansion, advanced AI models, real-time speech processing, multilingual support, interactive therapy features, and cloud-based analytics to further enhance the effectiveness and accessibility of the system

3.4 Conclusion

This research demonstrates the successful development and implementation of a Real-Time Stuttering Disorder Detection System using machine learning and speech signal processing techniques. The system analyzes speech recordings to detect stuttering patterns, classify severity levels, and provide therapy recommendations through a web-based interface.

Testing results showed that the machine learning model achieved approximately 93% accuracy in stuttering detection, over 85% accuracy in severity classification, and **87%** alignment with expert therapy recommendations. The web application interface enables users to easily upload speech samples, receive real-time analysis, and track their speech improvement.

The system also achieved a high usability score (SUS = 84/100), indicating that it is accessible and user-friendly for both individuals and speech therapists. By providing automated analysis and guidance, the system can support early detection of stuttering disorders and assist in speech therapy interventions.

In conclusion, this project highlights how machine learning, speech processing, and web technologies can be combined to create an effective tool for speech disorder detection and therapy support. The proposed system offers a cost-effective, scalable, and practical solution that can assist individuals in improving their speech fluency and enable therapists to monitor patient progress more efficiently.

With future enhancements such as advanced deep learning models, larger datasets, and real-time speech feedback, the system has the potential to become a comprehensive digital assistant for speech therapy and communication improvement.

3.4 References

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